

ERIC (CHEN-CHIEN) CHANG

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SUMMARY

- Backend / Data Infrastructure Engineer focused on **Kafka**, **PostgreSQL**, **distributed systems**, and **performance tuning**; delivered wins including first-byte latency **100s to 5s**, WebUI load time **60s to 1s**, and simulation jobs **2hr to 1min**.
- Core stack: **Go**, **Java**, **Rust**, **TypeScript**, **PostgreSQL**, **Kafka**, **ClickHouse**, **Cloudflare R2**, **Docker**, **Kubernetes**, **Airflow**, **Prometheus/Grafana**.

OPEN SOURCE CONTRIBUTIONS

- **Apache Kafka Contributor**: collaborating on Kafka Streams **KIP-770** input-buffer byte bounds ([PR #22458](#)) and fixed offset commit-before-partition-switch correctness ([PR #21584](#)).
- **Apache Gravitino Committer**: added PostgreSQL/MySQL integration-test container management, reproducible Docker image workflows, and reduced Hive image size by **420MB**.
- **Apache DataFusion / Comet Contributor**: implemented Spark-compatible `str_to_map` in Rust and wired Comet's native SparkStrToMap UDF with SQL coverage.
- **redis/rueidis Contributor**: implemented the go-redis API adapter for migration compatibility.

SELECTED PROJECTS

TWFoundry

2026

Hybrid cloud / homelab Taiwan operations platform. Tech Stack: [Airflow](#), [Kafka](#), [ClickHouse](#), [Cloudflare R2](#), [Workers/Pages](#), [Docker](#)

[GitHub](#)

- Designed a data platform using Airflow/Kafka for ingestion and backfills, R2 for raw and projection artifacts, ClickHouse for OLAP analytics, and Workers / Pages for edge-served map/timeline APIs without homelab fallback on cache misses.

EXPERIENCE

Lawsnote

January 2024 - December 2024

Backend Software Engineer. Tech Stack: [Typescript](#), [NodeJS](#), [Docker](#), [MongoDB](#), [Prometheus](#), [Grafana](#)

Taipei, Taiwan

- Implemented **HTTP streaming** for CPU-intensive computation results, reducing first-byte latency from **100s** to **5s**.
- Built a **MinIO events** and **Elasticsearch** monitoring path to detect missing data proactively and improve pipeline reliability.
- Built an **LLM service** for extracting information from ambiguous regulatory text and automated government act crawling, re-crawl, and re-parse workflows.

Mediatek (Contract)

February 2022 - March 2023

Backend Software Engineer. Tech Stack: [Go](#), [Linux](#), [Docker](#), [PostgreSQL](#)

Hsinchu, Taiwan

- Maintained a low-power simulation and benchmarking data warehouse, improving WebUI page load time from **60s** to **1s** with **PostgreSQL index-only scans**: [Blog Post](#).
- Used **Pyroscope eBPF** to identify import bottlenecks, then applied **PostgreSQL bulk import** to reduce simulation job duration from **2hr** to **1min**.